

AGENT5G

Agentic AI for 5G Advanced

An autonomous operations platform for 5G Advanced Release 20

5G Advanced · Release 20 | Project Showcase · 2026

THE PROBLEM

- **Exploding complexity**
Dense cells, slicing, RAN, Core and Edge create millions of interacting parameters no team can tune by hand.
- **Reactive operations**
Faults are found after they hurt users; diagnosis and remediation stay slow and human-dependent.
- **Rising OPEX & energy cost**
Round-the-clock monitoring and over-provisioning inflate operating and energy budgets while SLAs tighten.
- **Siloed tooling**
Metrics, logs, topology and planning live in disconnected tools with no unified intelligence layer.

THE SOLUTION

Agent5G observes, reasons and acts on the live 5G network through a closed automation loop - turning reactive operations into autonomous ones.

- **Autonomous agents**
Goal-driven agents detect, diagnose and resolve issues without waiting for a human.
- **Reasoning + memory**
LLM reasoning grounded by a knowledge graph and long-term memory of past incidents.
- **Digital twin safety**
Every action is validated on a digital twin before it ever touches the live network.

KEY FEATURES

Live Dashboard · Analytics · Agent Console · Workflow Builder · Memory · Knowledge Graph · Topology · Digital Twin · Simulation · Model Manager · Service Registry · Logs & Observability

HOW IT WORKS

- 1 **Observe**
Ingest live telemetry, metrics, logs and events.
- 2 **Reason**
Agents analyse with LLM + knowledge graph + memory.
- 3 **Plan**
Generate a remediation or optimisation plan.
- 4 **Simulate**
Validate the plan on the digital twin first.
- 5 **Act**
Execute the approved change autonomously.

6 Learn

Store the outcome and improve future decisions.

TECHNOLOGY STACK

Frontend

Next.js 16 · React 19 · TypeScript · Tailwind CSS v4 · Zustand · React Query · Recharts · React Flow · Framer Motion

Backend

Python · FastAPI · WebSockets · OpenAPI · Async I/O

AI & Intelligence

LLM agents · RAG memory · Knowledge graph · Digital twin · Simulation

Platform & Ops

Railway · Nixpacks · Playwright · Vitest

AI FEATURES

- **Multi-agent orchestration**
Specialised agents collaborate toward operational goals.
- **Natural-language ops**
Ask the network questions and issue intents in plain English.
- **Retrieval-augmented memory**
Past incidents inform every new decision via RAG.
- **Knowledge-graph reasoning**
Grounded, explainable decisions instead of black-box guesses.
- **Predictive analytics**
Anomaly detection and forecasting head off failures early.
- **Twin-verified safety**
Actions are proven safe in simulation before execution.

FUTURE SCOPE

- **O-RAN & multi-vendor RAN integration**
- **Reinforcement learning for self-improving optimisation**
- **Energy-aware green networking**
- **Intent-based networking (language to policy)**
- **Federated edge agents**
- **6G readiness**

TEAM

Divyansh Jaiswal

202311030 · Solo Developer - Full-stack & AI

Mentor: Dr. Bhupendra Kumar

Indian Institute of Information Technology, Vadodara (IIIT Vadodara ICD)